



ElectronicsTutorials

DIODE RECTIFICATION

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and Beyond

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TABLE OF
CONTENTS

1. Diode Rectification	1
2. The PN-junction Diode	1
3. Rectifier Circuits	2
4. The Half-wave Rectifier Circuit	2
5. Half-wave Ripple Factor	3
6. A Full-wave Rectifier Circuit	5
7. The Bridge Rectifier Circuit	7
8. Dual Supply Bridge Rectifier Circuit	9
9. Alternative Bridge Rectifier Layout.	10
10. Half-wave Three-phase Diode Rectification	10
11. Full-wave Three-phase Diode Rectification	12

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1. DIODE RECTIFICATION

DC voltage supplies are commonly produced from batteries, dynamos, and solar photovoltaic cells, etc. But we can also produce a DC (direct current) supply from an alternating current or AC supply using semiconductor diodes. The process of converting an AC supply into a DC supply is called **Rectification**.

We remember that a mains powered alternating current supply is one which flows for some part of its time in one direction and then, after a given time period, reverses itself, and then flows backwards in the opposite direction. This forwards and backwards movement of current (electrons) constitutes one full cycle, where the time taken to complete one full cycle being known as the periodic time, or period (T).

Then an alternating current (AC) supply is one which continually changes direction with the number of times it repeats itself in one second, commonly known as its frequency. The standard unit given for frequency is the Hertz (Hz). That is the number of Hertz represents the number of cycles (alterations) within one full second.

While an alternating current supply can have many different shapes, called waveforms, the most common waveform shape used for rectification purposes is that of the sine wave as they alternate periodically between some maximum and minimum value over one full cycle of the waveform.

However, batteries and DC supplies provide a direct current supply which always flows in one definite direction and also of a constant value. For example, voltages of: +5V, +12V, -6V, -9V, etc.

DC flows in one direction only. AC alternates between a positive and negative value each and every cycle

As almost all electronic circuits and equipment operate using direct current supplies, we need a means of converting this constantly changing bidirectional alternating current (or voltage) into a steady and constant unidirectional direct current (or voltage) supply.

One very simple, cheap and effective way of achieving this conversion is by using a semiconductor solid-state device called a pn-junction diode which acts like a one-way valve to the flow of electrons through it.

2. THE PN-JUNCTION DIODE

The PN-junction diode is a simple semiconductor device that has the characteristic of passing electrical current through itself in one direction only. The diode is formed by joining together two pieces of semiconductor material. One of these being a *n-type* material, and the other a *p-type* material.

The joining together of these two semiconductor materials results in the diode having a positive “p-region”, called the Anode (a) at one end, and a negative “n-region”, called the Cathode (k) at the other end. This Positive-Negative-Junction is the basic structure of all diodes and hence the name PN-junction diode.

FIGURE 1. SEMICONDUCTOR DIODE IV CHARACTERISTICS CURVE

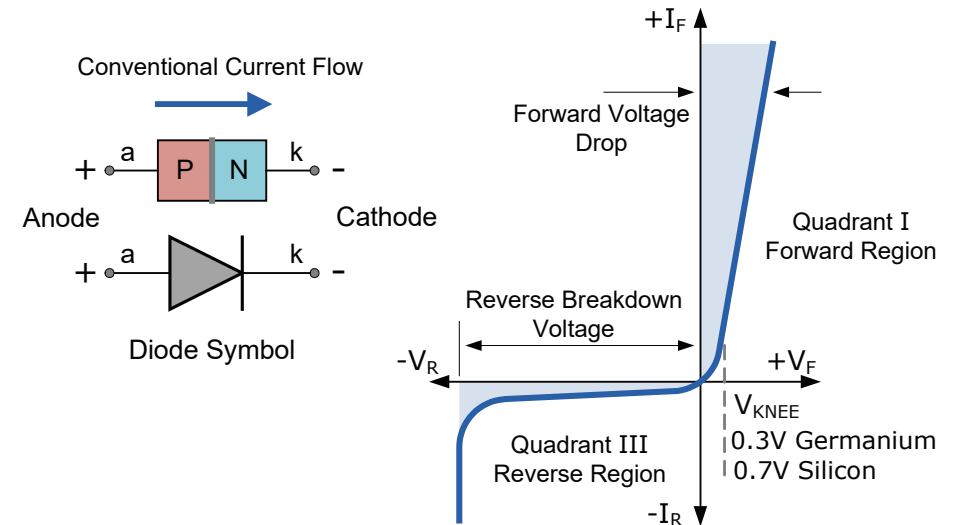


Figure 1. shows the typical construction, symbol and voltage-current (I-V) characteristics curve of a semiconductor pn-junction diode. The arrow shape of the diodes symbol represents the directional flow of conventional current through the diode from anode (a) terminal to its cathode (k) terminal.

When the anode terminal is made more positive with respect to the cathode terminal be more than 0.7 volts (or 0.3 volts for germanium semiconductor) the diode offers little resistance and allows current to easily flow through it. However, when the cathode is made more positive than the anode by any amount within its reverse breakdown region, it offers a very high circuit resistance preventing or stopping the flow of current through it.

So we can think of a semiconductor diode as being a sort of voltage sensitive switch that is closed or “ON” when the anode is more positive (by 0.7 volts), and open or “OFF” when the cathode is more positive (any voltage value). Clearly a perfect diode would have zero resistance when “ON” (conducting) and infinite resistance when “OFF” (blocking).

Then we can see that when the alternating current supply is positive in value it will flow through the diode, but when it is negative in value it is blocked by the diode. We can use this inbuilt characteristic of the diode to rectify a constantly changing bidirectional alternating current supply into a positive unidirectional direct current supply.

3. RECTIFIER CIRCUITS

We now know that rectifiers are used to convert an incoming AC voltage supply into DC voltage supply and are therefore the main component of any rectifier circuit. The AC incoming supply can be either a single-phase supply, a three-phase supply, or a poly-phase supply consisting of 6, 12 or more phases.

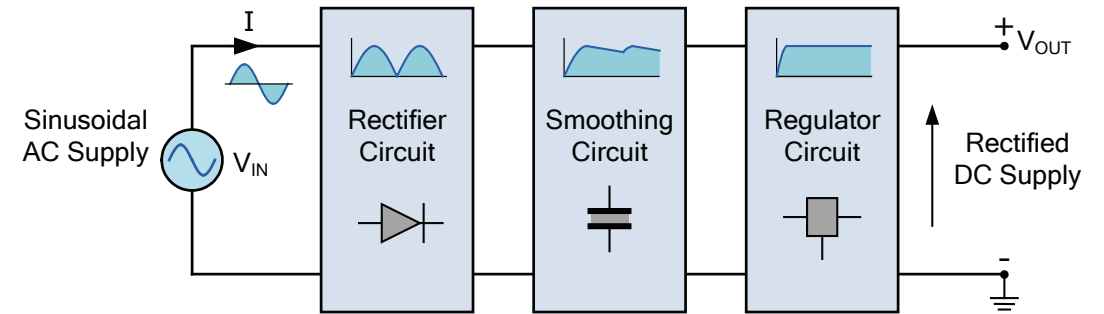
But these rectifier circuits on their own do not produce a steady-state DC supply such as those given by a battery or solar photovoltaic cell, but instead produce series of positive or negative going pulses, depending on which way around the rectifying diodes are connected within a circuit.

Then rectifiers produce what is called a pulsating DC supply, and this rough pulsating DC supply is then smoothed and condition by some filter circuit. Also variations in load current can be stabilised by using a suitable voltage regulator circuit connected to the rectifiers output.

The basic arrangement of a typical DC power supply is shown in Figure 2 where alternating

current enters and leaves as a smoothed and regulated DC supply.

FIGURE 2. A TYPICAL DC POWER SUPPLY ARRANGEMENT

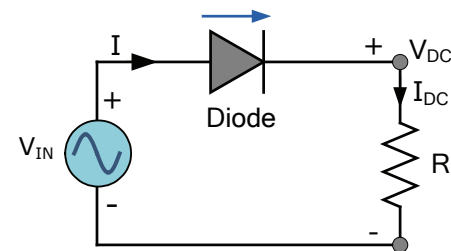


Generally, the input sinusoidal voltage is supplied by a suitable transformer which steps down the higher AC mains voltage to the desired voltage level required by the rectifier. The use of a transformer also provides electrical isolation from the mains AC supply to the circuit or system being supplied by the rectifier circuit.

4. THE HALF-WAVE RECTIFIER CIRCUIT

There are many different rectifier circuit designs containing diodes for converting an alternating AC current into a direct DC current supply, each with its own set of advantages and disadvantages.

FIGURE 3. HALF-WAVE RECTIFIER CIRCUIT



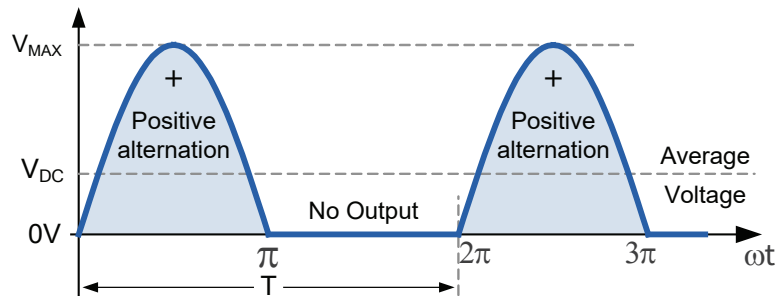
The most basic and simplest rectifier circuit is that of the single-phase half-wave unregulated rectifier which uses just one semiconductor diode within its design as shown in Figure 3.

The half-wave rectifier circuit of Figure 3. is so called because it uses only one-half of each full cycle of the AC supply.

This is because the diode conducts during positive half-cycles of input voltage when the diode is forward-biased, and cuts off during negative half-cycle when it is reverse-biased.

Then the output from a half-wave rectifier is basically a series of positive going pulses. These pulses have a frequency of the same value as the input supply frequency. If the input to the diode rectifier is sinusoidal, then the output waveform supplied to the resistive load R_L is shown in Figure 4.

FIGURE 4. HALF-WAVE RECTIFIER WAVEFORM



Then we can clearly see that when the diode is forward-biased during the positive half-cycle, there is an output voltage supplied to the load. When the diode is reverse-biased during the negative half-cycle, it acts as an open-circuit so the output voltage is zero during this time and no current flows in the load resistor.

However, when the input voltage waveform goes negative and the diode becomes reverse-biased, all of the input voltage V_{IN} appears directly across the diode so the peak inverse voltage (PIV) of the diode must be higher to prevent breakdown.

Then the voltage developed across the load resistor is a series of half-cycle positive pulses, hence the name half-wave rectifier. As the peak or maximum voltage value of the output waveform is equal to the peak or maximum value of the sinusoidal input waveform (minus one 0.7V diode voltage drop).

Since no current flows during the negative half-cycle of the AC input voltage, we can calculate the average value of the output over one half a cycle from 0 to π (0 to 180°).

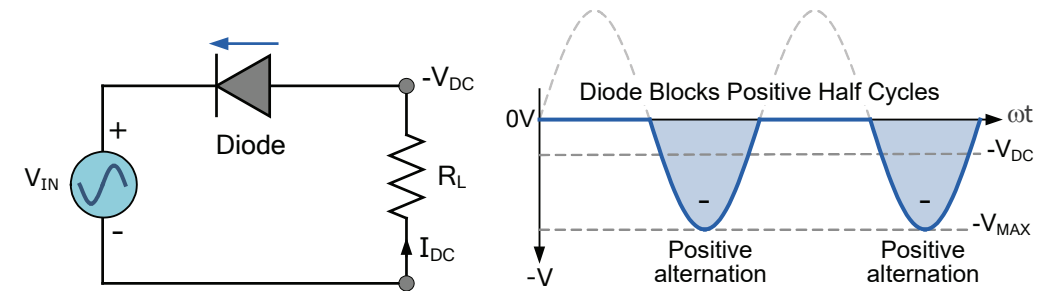
Then we can calculate the average DC voltage value supplied to the load resistor as being:

$$V_{DC} = \frac{V_{MAX}}{\pi} = \frac{1}{\pi} V_{MAX} = 0.318 V_{MAX} \text{ or } 0.45 V_{RMS}$$

Where: V_{MAX} is the maximum or peak value of the sinusoidal waveform, and V_{RMS} is its root mean squared value of the voltage and where $V_{RMS} = 0.7071 V_{MAX}$. The DC current flowing through the resistive load is always in one positive direction in spite of it being a pulsating DC supply. The load current is therefore given as: $I_{DC} = V_{DC}/R$.

By reversing the direction of the rectifying diode, we can produce a negative pulsating supply as shown in Figure 5. The diode is only forward-biased during the negative half cycles of the AC supply waveform and so blocks ALL the positive half-cycles.

FIGURE 5. HALF-WAVE NEGATIVE RECTIFICATION



Then the polarity of the output voltage for a half-wave rectifier depends on which way around the diode is connected within the half-wave rectifier circuit.

5. HALF-WAVE RIPPLE FACTOR

We have seen that the half-wave rectifier circuit produces a pulsating DC output voltage that has an average DC output value of V_{DC} . But this pulsating DC output is not suitable for powering electronic circuits that require a steady-state DC supply, such as that obtained from batteries.

The Ripple Factor (RF) is another parameter used to describe the quality of the rectifiers output waveform. As we want the output voltage and current as smooth as possible, the ripple factor represents the smoothness of the voltage waveform.

Mathematically, this ripple voltage can be defined by a ripple factor which expresses how smooth the DC output is. Ripple factor is defined as the ratio of the sinusoidal RMS value to the DC output value. That is: $V_{AC(RMS)}/V_{DC}$ and we can define this half-wave rectifier ripple factor as being:

$$V_{DC} = \frac{V_{MAX}}{2}, \text{ and } V_{RMS} = \frac{V_{MAX}}{2}$$

$$\text{Ripple Factor} = \frac{V_{AC(RMS)}}{V_{DC}} = \frac{\sqrt{V_{RMS}^2 - V_{DC}^2}}{V_{DC}}$$

Thus:

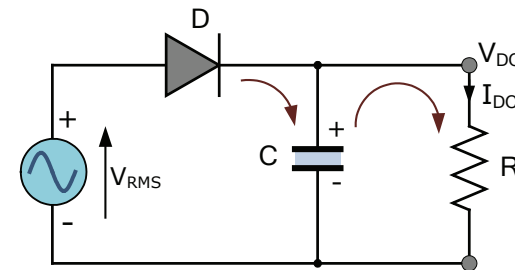
$$\text{Ripple Factor} = \sqrt{\left(\frac{V_{RMS}}{V_{DC}}\right)^2 - 1} = \sqrt{\left[\frac{\left(\frac{V_{MAX}}{2}\right)}{\left(\frac{V_{MAX}}{2}\right)}\right]^2 - 1} = \sqrt{\frac{1}{4} - 1} = \sqrt{1.468} = 1.211$$

Then the ripple factor for an unsmoothed half-wave rectifier is given as being: **1.211**. That is 1.211 times greater than the DC component, which is not good.

Then we need to add an additional component to smooth out these pulsating peaks called “ripples” (either positive or negative) which are the same amplitude and shape of the sinusoidal supply to improve the average DC value of the output waveform and get closer to pure DC.

We can do this by simply adding a smoothing circuit in the form of a reservoir capacitor to the rectifiers output as shown in Figure 6.

FIGURE 6. HALF-WAVE RECTIFIER SMOOTHING CIRCUIT



During the positive half-cycle, diode D conducts charging the capacitor, C to the peak voltage value of the supply. During the negative half-cycle, the diode is reverse biased so capacitor C slowly starts to discharge through the load resistor R_L .

The capacitor continues to discharge through the load resistor as the supply voltage

reduces until once again the diode gets forward biased and starts conducting on the next positive pulse. The capacitor charges up again until the peak voltage value is reached and the cycle continues.

This constant charging and discharging of the reservoir capacitor every pulse results in a smoother ripple voltage at the output as shown in Figure 7.

FIGURE 7. HALF-WAVE SMOOTHING CAPACITOR WAVEFORM

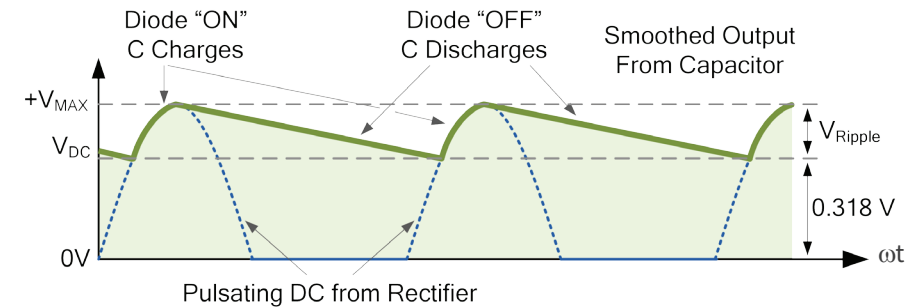


Figure 7. shows that the constant charging and discharging of the capacitor results in a flatter shaped voltage waveform with less ripple at the output. Thus a larger value capacitor, and/or load resistance, R_L will result in a slower discharge rate due to an increase in RC time constant and therefore, a more flattened output waveform with less ripple content.

A half-wave rectifier with a smoothing capacitor results in a large RC time constant, this ripple factor is given by:

$$\text{Ripple Factor} = 1.211 = \frac{1}{2\sqrt{3}fCR_L}$$

Where f is the frequency of the rectified signal, C is the value of the smoothing capacitor in farads, and R_L is the resistive value of the connected load. Then clearly we can see that the value of the ripple voltage will decrease if either f , C or R_L increases as the capacitor discharges less. Note that the product of CR_L gives the time constant of the circuit.

Although half-wave rectifiers can be used in battery charging or signal conditioning applications, as we have seen, their main disadvantage is that there is no output voltage for half of the available input waveform making it quite inefficient. Also their regulation is poor due to their high ripple factor of 1.211.

One way to improve on this design and increase the output voltage while decreasing the ripple factor is to add a second diode to produce what is commonly called a full-wave rectifier circuit.

6. A FULL-WAVE RECTIFIER CIRCUIT

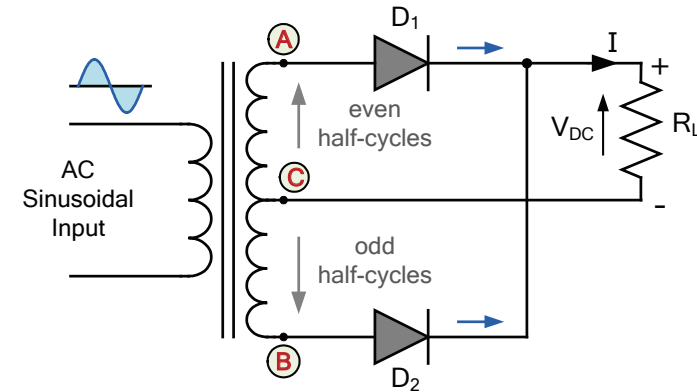
As its name implies, a full-wave rectifier circuit uses both the positive and negative half-cycles of the sinusoidal input waveform to produce a DC supply. The full-wave rectifier is effectively two half-wave rectifiers connected together which take it in turn to supply current to the load through two rectifying diodes as shown in Figure 8.

The circuit of Figure 8. consists of a center-tapped transformer, and a rectifier formed by two diodes D_1 and D_2 . The transformer's secondary winding is split equally into two halves with a center tap connection, (C). This acts as a common neutral or zero point.

On the positive half-cycle of the AC input waveform, point A of the transformer's upper secondary winding becomes more positive with respect to point C, diode D_1 therefore conducts in the forward direction delivering current to the load, ($I_L = I_{D1}$).

As point B of the transformer's lower secondary winding is negative with respect to point C at this time, diode D_2 is reverse biased and cut-off.

FIGURE 8. FULL-WAVE RECTIFIER CIRCUIT



On the negative half-cycle of the AC input waveform the reverse occurs. Point B of the transformer's lower winding now becomes more positive with respect to the common point C. Thus diode D_2 conducts in the forward direction supplying load current, ($I_L = I_{D2}$) as diode D_1 is now reverse biased and cut-off.

The current flowing through the load resistor R_L is therefore the same direction for both half-cycles as each diode takes it in turn to supply current to the load. The empty space in between the half-cycles previously seen in the half-wave rectifier circuit is now filled in by the second diode supplying a fully rectified output waveform to the load.

Then the two diodes feed current to the resistive load, R_L from the transformer. The result of this centre-tapped full-wave rectifier is an output voltage with a ripple frequency twice the input frequency and that pulsates every half-cycle of the input waveform.

Clearly, the centre-tapped transformer must be of a suitable power rating and both half windings equal in turns.

The half-wave rectifier uses only one diode. The full-wave rectifier uses two diodes

FIGURE 9. FULL-WAVE RECTIFIER WAVEFORM

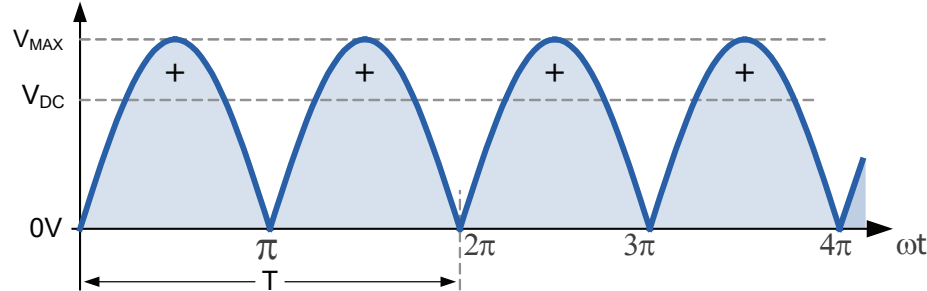


Figure 9. shows that the peak or maximum voltage, V_{MAX} developed across the load resistor R_L is therefore unidirectional and of the same value as the input waveform (minus one 0.7V diode forward voltage drop). However, as we now have two diodes alternatively supplying current to the resistive load, the load current I_L is therefore the sum of the two individual diode currents.

The output load voltage from a full-wave rectifier is still pulsating DC and not pure steady-state DC, but average DC output voltage across the load resistor, R_L is now double that of the previous single half-wave rectifier circuit providing each half of the transformer windings have the same RMS voltage value. This is because the average DC load voltage is now calculated from 0 to 2π (0 to 360°).

If for the previous half-wave rectifier circuit, $V_{DC} = V_{MAX}/\pi$ then as the full-wave rectifier circuit supplies two half-cycles to the load, then for a full wave rectifier the average DC voltage value supplied to the load resistor is:

$$V_{DC} = \frac{2V_{MAX}}{\pi} = \frac{2}{\pi} V_{MAX} = 0.636 V_{MAX} \text{ or } 0.9 V_{RMS}$$

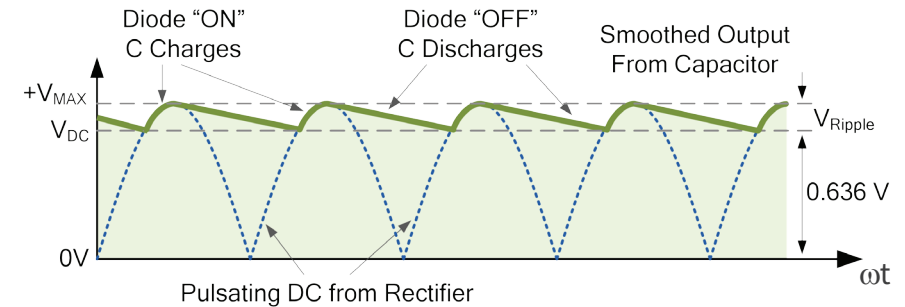
Where again, V_{MAX} is the maximum or peak value of the sinusoidal waveform, and V_{RMS} is its RMS value of the voltage as: $V_{RMS} = 0.7071 V_{MAX}$. The DC current is given as: $I_{DC} = V_{DC}/R_L$.

For the centre-tapped full-wave rectifier $V_{DC} = 2V_{MAX}/\pi$, the ripple factor is calculated using the generalised formula above as being: **0.482**. While this ripple factor value is greatly

reduced from the previous **1.211** value of the half-wave rectifier circuit, it is still pulsating DC with a large AC component.

As before, a smoothing capacitor is still required at the rectifiers output to smooth out these pulsating ripples with the same idea applied to the full wave rectifier circuit. The difference this time is that the time required by the capacitor, C to charge upto V_{MAX} is now smaller and only discharges through the load resistor when the two diodes are not conducting as shown in Figure 10.

FIGURE 10. FULL-WAVE CAPACITOR FILTER WAVEFORM



Then we can see from Figure 10 that as the charging and discharging times are small due to the large RC time constant, the ripple voltage is also small. Thus the ripple factor given for a two diode full-wave rectifier circuit with smoothing capacitor is given as:

$$\text{Ripple Factor} = 0.482 = \frac{1}{4\sqrt{3}fCR_L}$$

Where f is the frequency of the rectified signal, C is the value of the smoothing capacitor in farads, and R_L is the resistive value of the connected load. So assuming an AC supply frequency or 50Hz, and a 2kΩ load, a 100uF capacitor would reduce the ripple voltage to only 14.4mV. Again, the product of CR_L gives the time constant of the circuit.

Full-wave rectifiers supplying a heavy current load with a single smoothing or reservoir capacitor can be improved by the using a large capacitance value, as it stores more energy and discharges less between charging pulses.

Generally, for full-wave DC power supply circuits the single smoothing capacitor is an Aluminium Electrolytic type which has a capacitance value of more than 100uF. Inductive-Capacitive (LC) pi-filters can also be used.

The main disadvantage of the previous single-phase full-wave rectifier circuit is that a center-tapped secondary transformer is required, or one with two separate but identical secondary windings to provide the equal plus and minus voltages. This therefore makes this type of full wave rectifying circuit costly.

While this full-wave rectifier circuit is more efficient than the half-wave rectifier circuit as it uses both halves of the input waveform, we can improve on this by adding two more rectifying diodes to create a “Full Wave Bridge Rectifier” circuit equivalent.

7. THE BRIDGE RECTIFIER CIRCUIT

The bridge rectifier is another type of single-phase full-wave rectification circuit that uses both half-cycles of the input waveform to produce a pulsating output voltage. However, the bridge rectifier circuit uses four individual rectifying diodes connected in a closed loop “bridge” configuration to produce the desired output.

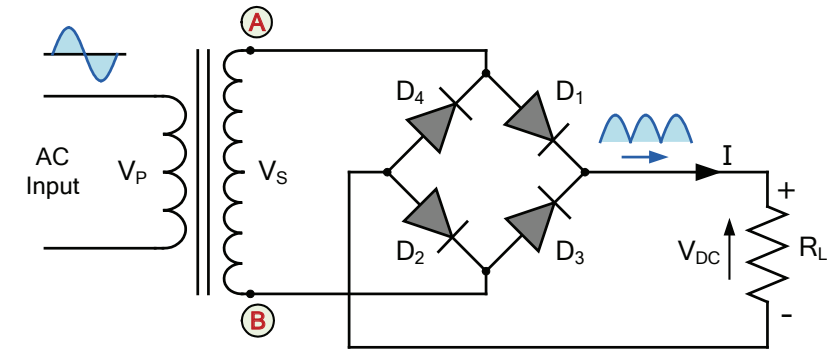
The big advantage of the bridge rectifier circuit is that it does not require a special center-tapped secondary transformer to operate, in fact it can operate with or without a transformer as an AC supply can be connected directly to the bridge.

If a transformer is used, it requires only one single secondary winding with an appropriate turns ratio ($N_p:N_s$), thereby reducing its size and cost. Also, the transformer can be used to step-down or step-up the supply voltage as required, as well as providing electrical isolation between the AC input and DC output supplies.

In the circuit of Figure 11. the voltage supply from the transformers secondary winding is connected to one side of the diode bridge network and the resistive load to the other side as shown.

The bridge rectifier uses four interconnected diodes to provide rectification

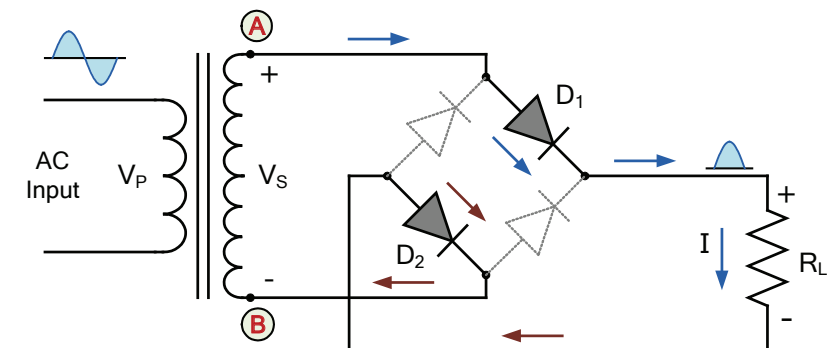
FIGURE 11. DIODE BRIDGE RECTIFIER CIRCUIT



The four rectifying diodes, labeled D_1 to D_4 are arranged as “series pairs” within a bridge configuration. The four diodes are so arranged so that the current can only flow in only one direction through the load with only two diodes conducting current at any time during each half cycle.

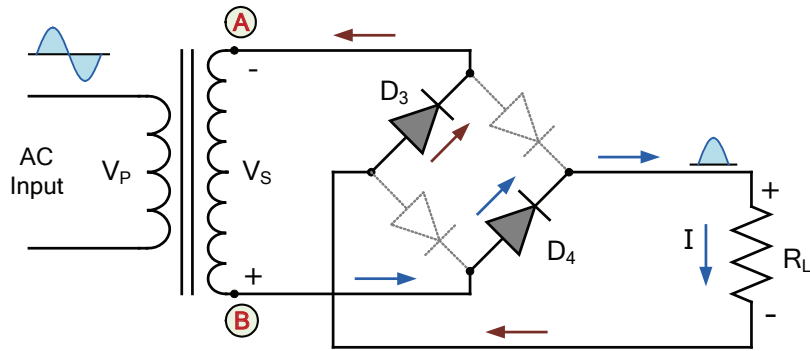
The operation of bridge rectifier circuit is fairly straightforward. During the positive half-cycle of the secondary supply V_S , point A of the secondary winding becomes positive with respect to point B. Diodes D_1 and D_2 become forward biased and conduct in series while diodes D_3 and D_4 are reverse biased and cut-off. Therefore, current flows through the load from point A and through D_1 R_L D_2 and back to point B as shown in Figure 12.

FIGURE 12. CURRENT FLOW DURING POSITIVE HALF-CYCLE



During the negative half-cycle of the secondary supply V_s , point B of the secondary winding now becomes more positive than point A. Diodes D_3 and D_4 are now forward biased and conduct as a series pair. Diodes D_1 and D_2 become reverse biased and so are cut-off. Therefore, the current flows from point B and through D_3 , R_L , D_4 and back to the secondary winding at point A as shown in Figure 13.

FIGURE 13. CURRENT FLOW DURING NEGATIVE HALF-CYCLE



Then no matter which half cycle is being supplied by the secondary of the transformer, the current, I flowing through the load is always in the same direction producing a full-wave rectified output.

As the current flowing through the resistive load, R_L is unidirectional, so the voltage developed across the load is also unidirectional the same as for the previous two diode full-wave rectifier.

Therefore, the average DC voltage developed across the load resistance will be exactly the same. Thus the corresponding expressions for the average DC value of the voltage or current for a full-wave bridge rectifier is given as:

$$V_{DC} = 0.637 \times V_{MAX}$$

$$I_{DC} = 0.637 \times I_{MAX}$$

As before, the maximum value is the peak value of the input waveform but we could also use its RMS, or root mean squared value to find the equivalent DC output value of a single phase full-wave rectifier. To determine the average voltage for a full-wave bridge rectifier, we must multiply the waveforms RMS value by 0.9 giving:

$$V_{DC} = 0.9 \times V_{RMS}$$

$$I_{DC} = 0.9 \times I_{rms}$$

Then we can see that a full-wave bridge rectifier circuit converts BOTH the positive or negative halves of an AC waveform into a pulsed DC output that has a value of $0.637V_{MAX}$ or $0.9V_{RMS}$.

However in reality, during each half cycle the current flows through two diodes instead of just one so the amplitude of the DC voltage output is two diode voltage drops ($2 \times 0.7 = 1.4V$) less than the input V_{MAX} amplitude.

The ripple frequency is twice the supply frequency (e.g. 100Hz for a 50Hz supply or 120Hz for a 60Hz supply) as before. The output voltage waveform can be smoothed again by connecting a suitable capacitor across the load resistor, R_L .

One small disadvantage is that the bridge rectifier requires four semiconductor diodes to operate. However, with no large centre-tapped transformer required compared to the previous full-wave rectifier, and as rectifying diodes such as the 1N4007 series costing only a few pennies, the overall cost of a bridge rectifier is much cheaper.

Although we can use four individual power diodes to make a full wave bridge rectifier circuit, pre-made bridge rectifier components are available "off-the-shelf" in a range of different voltage and current ratings to suit all applications.

These rectifying devices can be soldered directly onto a PCB circuit board or have spade connectors for external cables. Thus full-wave bridge rectifiers are more commonly used than half-wave or full-wave rectifiers using individual diodes.

TABLE 1. COMPARISON OF DIODE RECTIFIER CIRCUITS

	Parameter	Half-wave	Full-wave	Bridge
1	Number of Diodes Used	1	2	4
2	Average DC Voltage (V_{DC})	$\frac{V_{MAX}}{\pi}$	$\frac{2V_{MAX}}{\pi}$	$\frac{2V_{MAX}}{\pi}$
3	Average DC Current (I_{DC})	$\frac{I_{MAX}}{\pi}$	$\frac{2I_{MAX}}{\pi}$	$\frac{2I_{MAX}}{\pi}$
4	RMS Current (I_{RMS})	$\frac{I_{MAX}}{2}$	$\frac{I_{MAX}}{\sqrt{2}}$	$\frac{I_{MAX}}{\sqrt{2}}$
5	Output Ripple Factor	1.211	0.482	0.482
6	Conversion Efficiency	40%	81%	81%

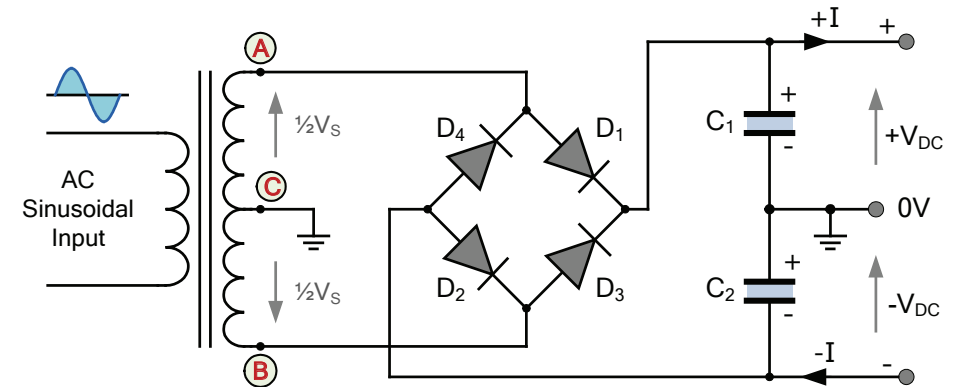
We can take the single-phase full-wave bridge rectifier circuit one step further to create a dual supply with both positive and negative voltage outputs.

8. DUAL SUPPLY BRIDGE RECTIFIER CIRCUIT

We said previously that the advantage of the bridge rectifier configuration is that it does not require a special center-tapped secondary transformer to operate. However, we can use a bridge rectifier with a center-tapped transformer to create a dual supply with two output voltages.

The center-tapped transformer used previously in the full-wave rectifier circuit of Figure 8, allows for two voltage phases to be produced from the secondary winding and which are 180 electrical degrees apart when the center tap, **C** is grounded as shown in Figure 14.

FIGURE 14. DUAL SUPPLY RECTIFIER CIRCUIT



The secondary of the transformer can consist of either one single centre-tapped winding, or two individual secondary windings connected together in series with the common centre connection grounded. For example, a transformer having two separate 9 volt secondary windings can be connected in series to provide a 9 – 0 – 9 (± 9) voltage output.

The two outputs from the bridge rectifier now become the positive and negative DC voltages referenced to ground. Two filter capacitors are connected in series between the positive and negative rails with the middle connection grounded to provide the required ripple smoothing. The resistive loads can therefore be connected between $+V_{DC}$ and $-V_{DC}$ rails, or between either supply rail and ground.

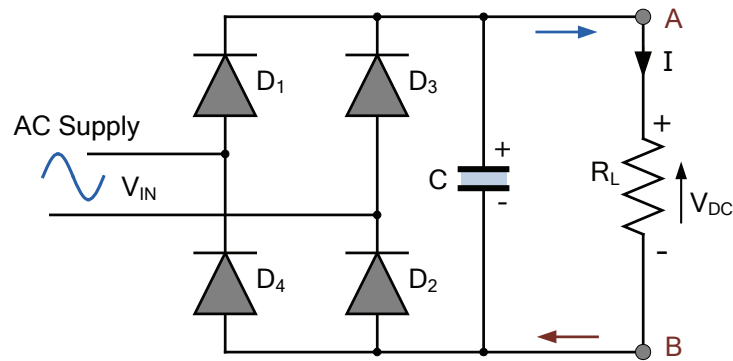
The advantage of a dual voltage supply is that can be used to power operational amplifiers, audio power amplifiers, bipolarity oscillators and range of filter circuits. Also, as fixed voltage regulators are available in both positive and negative versions, dual supplies can easily be created to supply a whole range of electronic circuits at the required DC voltage.

The most common types of three-terminal fixed voltage regulators are the 78... series positive regulators. Such as the 7805 (+5V), 7809 (+9V), 7812 (+12V), etc. and its complementary 79... series of negative regulators, 7905 (–5 V), 7912 (–12 V), etc.

9. ALTERNATIVE BRIDGE RECTIFIER LAYOUT

As well as the standard and common diamond shaped presentation of a full-wave diode bridge circuit, we can also present the 4-diode bridge rectifier as shown in figure 15.

FIGURE 15. ALTERNATIVE BRIDGE RECTIFIER CIRCUIT



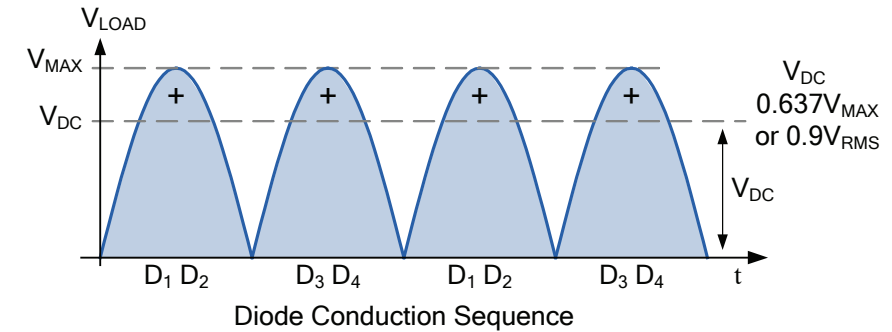
This series-parallel configuration of diodes still provides full-wave rectification because at any instant in time only two of the four diodes are forward biased while the other two are reverse biased the same as the previous diamond shaped configuration.

During the positive half cycle of V_{IN} , diodes D_1 and D_2 are forward biased while diodes D_3 and D_4 are reverse biased. So for the positive half-cycle of the input waveform, current flows along the closed path of: $D_1 - A - R_L - B - D_2$ and back to the supply.

During the negative half cycle of V_{IN} , diodes D_3 and D_4 are now forward biased while diodes D_1 and D_2 are reverse biased. Then for the negative half cycle of the input waveform, current flows along the closed path of: $D_3 - A - R_L - B - D_4$ and back to the supply.

Then the load current, I will always flow in the SAME direction through the load, R_L between terminals **A** and **B**. Thus whichever set of diodes are conducting, node **A** will always be more positive than node **B** producing a unidirectional DC supply with the unsmoothed output waveform shown in Figure 16.

FIGURE 16. FULL-WAVE BRIDGE RECTIFIER WAVEFORM



Then while the supply voltage may reverse at the beginning of each half-cycle, the voltage polarity across terminals **A - B** does not change. However, it does supply the total maximum (peak) voltage of V_{IN} and thus the average DC value is higher.

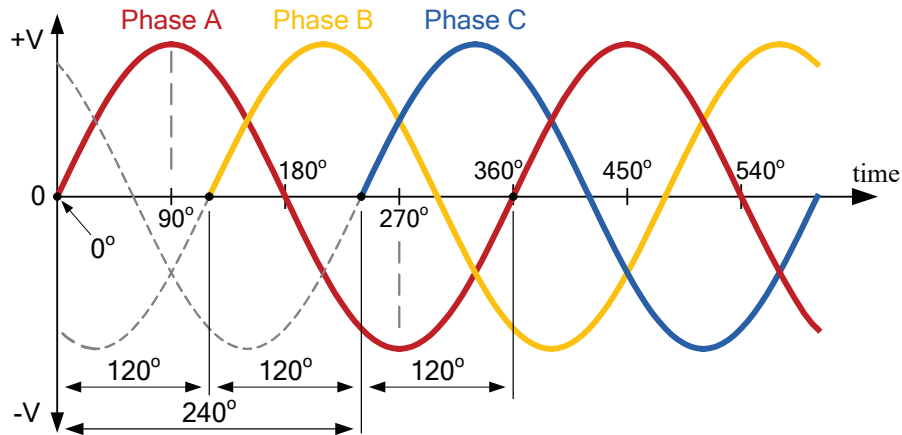
10. HALF-WAVE THREE-PHASE DIODE RECTIFICATION

Three-phase rectification, also known as poly-phase rectification circuits are similar to the previous single-phase rectifiers, the difference this time is that we are using three, single-phase supplies connected together that have been produced by one single three-phase generator in order to supply three rectifying diodes.

A 3-phase rectification circuit can be used to power many industrial applications such as DC motor control or battery charging systems which require a much higher power requirement than a single-phase rectifier circuit is able to supply.

3-phase supplies take this idea one step further by combining together three AC voltages of identical frequency and amplitude with each AC voltage being called a “*phase*”. These three phases are 120 electrical degrees “**out-of-phase**” from each other producing a phase sequence, or phase rotation of: $360^\circ \div 3 = 120^\circ$ as shown in Figure 17.

FIGURE 17. A THREE-PHASE VOLTAGE WAVEFORM



The advantage of a 3-phase AC supply is that it can be used to provide electrical power directly to balanced loads and diode rectifiers. Since a 3-phase supply has a fixed voltage and frequency it can be used by a rectification circuit to produce a pulsating DC voltage supply with less ripple compared to a single-phase rectifying circuit.

As with the previous single-phase rectifier, the most basic three-phase rectifier circuit is that of an uncontrolled half-wave rectifier circuit which uses three semiconductor diodes, one diode per phase as shown in Figure 18.

FIGURE 18. HALF-WAVE THREE-PHASE RECTIFIER CIRCUIT

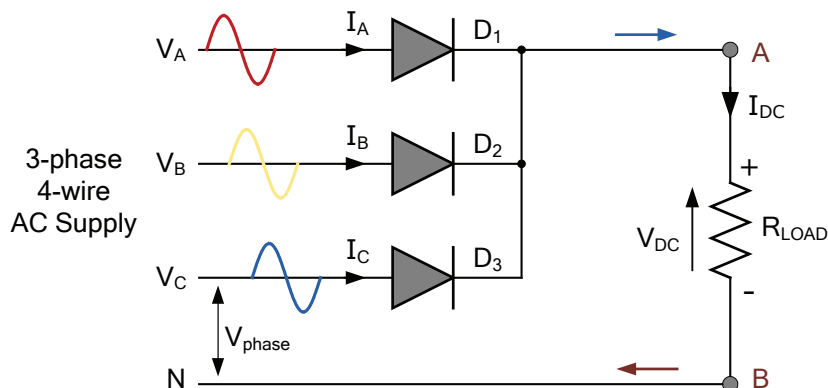


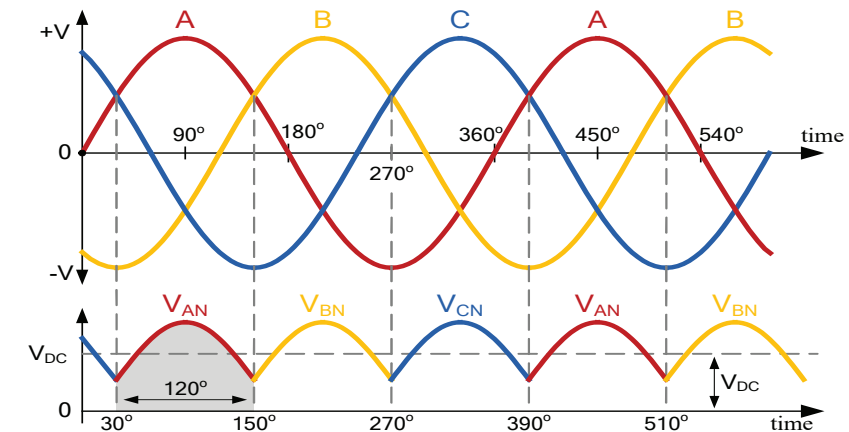
Figure 18 shows a 3-phase half-wave rectifier circuit. The anode terminal of each diode is connected to one phase of the voltage supply. The cathodes of all three diodes connected to the same positive point, effectively creating a “diode-OR” type arrangement. This common point, **A** becomes the positive (+) terminal for the load, while the negative (-) terminal at point **B** is connected to the neutral, **N** of the 3-phase, 4-wire supply.

Assuming a phase rotation of Red-Yellow-Blue ($V_A - V_B - V_C$) with the Red phase (V_A) starting at 0° . The first diode to conduct will be diode D_1 as it will have a more positive voltage at its anode than diodes D_2 or D_3 . Thus diode D_1 conducts for the positive half-cycle of V_A while D_2 and D_3 are in their reverse-biased state. The neutral (**N**) wire provides a return path for the load current back to the supply.

120 electrical degrees later, diode D_2 starts to conduct for the positive half-cycle of V_B (yellow phase) as now its anode terminal becomes more positive than those of diodes D_1 or D_3 . Thus load current switches from D_1 to D_2 . Similarly, another 120° later, V_C (blue phase) starts to increase turning “ON” diode D_3 as its anode becomes more positive, thus turning “OFF” diodes D_1 and D_2 .

Then we can see that for three-phase rectification, whichever rectifying diode has a more positive voltage at its anode terminal compared to the other two diodes, it will automatically start to conduct passing current to the connected load. This gives a conduction pattern of: $D_1 D_2 D_3$ as shown in Figure 19.

FIGURE 19. HALF-WAVE THREE-PHASE RECTIFIER CONDUCTION WAVEFORM



From the waveforms of Figure 19, for a resistive load, we can see that for a half-wave rectifier each diode passes current for only one third of each cycle. Thus the output waveform is three times the input frequency of the supply. Therefore, there are three voltage peaks in any given cycle.

By increasing the number of phases from a single-phase to a three-phase supply, the rectification of the supply is improved, that is the output DC voltage is much smoother as we now have 3-peaks and not one or two as before.

For a three-phase half-wave rectifier, the supply voltages V_A , V_B and V_C are balanced but with a phase difference of 120° . Thus the average DC value of the output voltage waveform from a 3-phase half-wave rectifier is given as:

$$V_{DC} = \frac{3\sqrt{3}}{2\pi} V_{MAX} = 0.827 V_{MAX}$$

As the voltage supplies peak or maximum voltage, V_{MAX} which is equal to $1.414 \times V_{RMS}$, it therefore follows that V_{RMS} is equal to $V_{MAX} \div 1.414$, or $0.707 \times V_{MAX}$ as: $1 \div 1.414 = 0.707$. Then the average DC output voltage of the rectifier can be expressed in terms of its root-mean-squared (RMS) phase voltage as being:

$$V_{DC} = 0.827 \times V_{MAX} = 0.827 \times 1.414 \times V_{RMS}$$

$$\therefore V_{DC} = \frac{0.827}{0.7071} V_{RMS} = 1.17 V_{RMS}$$

One of the disadvantages of half-wave 3-phase rectification is that it requires a 4-wire supply, that is three phases plus a neutral (N) connection. Also the average DC output voltage is low at a value represented by $0.827 \times V_{MAX}$ as we have seen. This is because the output ripple content is three times the input frequency.

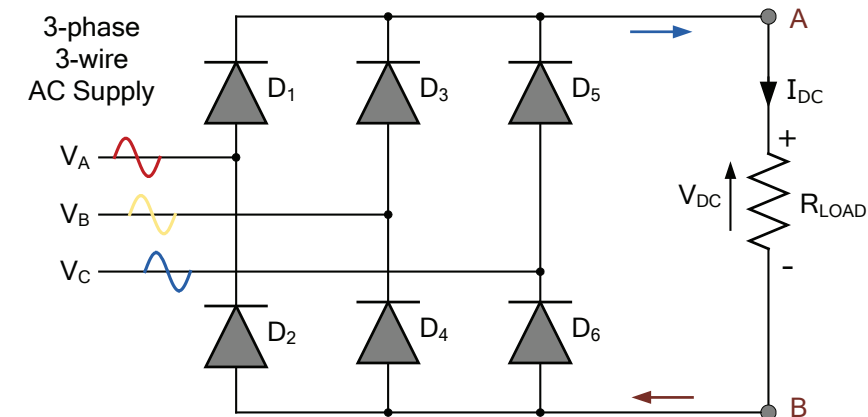
But we can improve on these disadvantages by adding three more diodes to the basic rectifier circuit creating a three-phase full-wave uncontrolled bridge rectifier.

11. FULL-WAVE THREE-PHASE DIODE RECTIFICATION

The full-wave three-phase uncontrolled bridge rectifier circuit uses six diodes, two per phase in a similar fashion to the single-phase bridge rectifier. A 3-phase full-wave rectifier is obtained by using two half-wave rectifier circuits. The advantage here is that the circuit produces a much lower ripple output than the previous half-wave 3-phase rectifier as it has a pulsating output frequency of six times the input AC waveform.

Also, the full-wave rectifier can be fed from a balanced 3-phase 3-wire delta connected supply as no fourth neutral (N) wire is required. Consider the full-wave 3-phase rectifier circuit shown in Figure 20.

FIGURE 20. FULL-WAVE THREE-PHASE RECTIFIER CIRCUIT



As before, assuming a phase rotation of Red-Yellow-Blue ($V_A - V_B - V_C$) and the red phase (V_A) starts at 0° . Each phase connects between a pair of diodes as shown. One diode of the conducting pair powers the positive (+) side of load, while the other diode powers the negative (-) side of the resistive load.

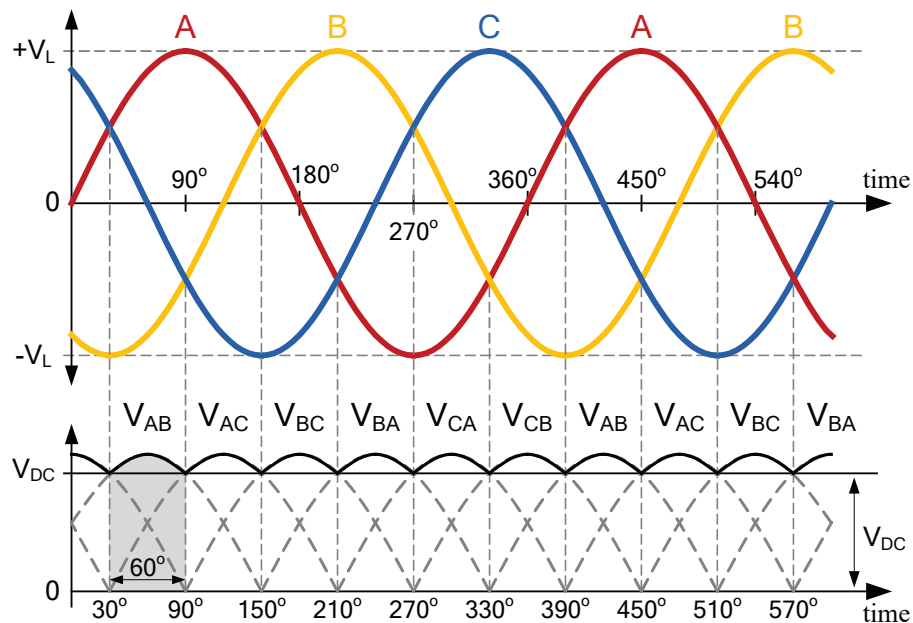
Diodes D_1 , D_3 , D_2 and D_4 form a bridge rectifier network between phases A and B, similarly diodes D_3 , D_5 , D_4 and D_6 between phases B and C and D_5 , D_1 , D_6 and D_2 between phases C and A. Diodes D_1 , D_3 and D_5 feed the positive rail.

The diode which has a more positive voltage at its anode terminal conducts. Likewise, diodes D_2 , D_4 and D_6 feed the negative rail and whichever diode has a more negative voltage at its cathode terminal conducts.

Then we can see that for three-phase uncontrolled rectification the diodes conduct in matching pairs with each conduction path passing through two diodes in series. Thus a total of six rectifier diodes are required for this configuration with commutation of the circuit taking place every 60° , or six times per cycle.

If we start the pattern of conduction at 30° , this gives us a conduction pattern for the load current of: D_{1-4} , D_{1-6} , D_{3-6} , D_{3-2} , D_{5-2} , D_{5-4} and return again to D_{1-4} and D_{1-6} for the next phase sequence as shown in Figure 21.

FIGURE 21. FULL-WAVE THREE-PHASE RECTIFIER CONDUCTION WAVEFORM



In 3-phase power rectifiers, conduction always occurs in the most positive forward-biased diode and the corresponding most negative diode. Thus as the three phases rotate across the rectifier terminals, conduction passes from diode to diode.

Each diode conducts for 120° (one-third) of each supply cycle but it takes two diodes to conduct in pairs. Each pair of diodes conduct for only 60° (one-sixth) of a cycle at any one time as shown in Figure 21.

Thus the average DC value of the output voltage waveform from a 3-phase full-wave rectifier is given as:

$$V_{DC} = \frac{3\sqrt{3}}{\pi} \times V_{MAX} = 1.654 V_{MAX}$$

or

$$V_{DC} = \frac{1.654}{0.7071} \times V_{RMS} = 2.34 V_{RMS}$$

Note also that unlike the previous half-wave rectifier, there is no common neutral connection between the rectifiers input and output terminals. Therefore, it can be fed by a balanced star (wye) or delta connected transformer supply.

[End of this Diode Rectification eBook](#)

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